The Association Between Daily Screen Time and Childhood Overweight and Obesity: Current Evidence and Potential Mechanisms

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Resumo (Abstract)

Abstract In recent decades, the widespread adoption of digital devices has led to a dramatic increase in screen time among children, raising significant public health concerns. This narrative review synthesizes current evidence on the relationship between daily screen exposure and the rising prevalence of childhood overweight and obesity. Childhood overweight and obesity are established risk factors for children's mental health problems in the short term and for the occurrence and development of various chronic diseases in adulthood. We searched PubMed, Web of Science, and Scopus for observational and intervention studies published in English between 2005 and 2026, using combinations of the terms "screen time", "television", "video games", "social media", "children", "adolescents", "overweight", and "obesity". Mounting evidence suggests that prolonged engagement with screens whether through television, smartphones, tablets, or computers is associated with to an elevated risk of weight-related health issues in children. However, the strength and consistency of this association vary across age groups, screen modalities, and study designs, with much of the evidence deriving from observational research that precludes definitive causal inference. Our review evaluates existing literature, encompassing recent empirical findings, theoretical perspectives, and proposed mechanistic pathways underlying this association. Key factors under examination include sedentary behavior displacing physical activity, exposure to unhealthy food marketing, disrupted sleep patterns, and the potential metabolic effects of prolonged screen use. Notably, these pathways are presented as proposed rather than definitively established mechanisms, given the predominance of cross-sectional and longitudinal observational data. Furthermore, this review explores evidence-based intervention strategies aimed at mitigating screen-related health risks, emphasizing the role of parental guidance, school-based policies, and public health initiatives in fostering healthier developmental trajectories for children. By integrating multidisciplinary insights, this review aims to inform future research directions and practical interventions to address this pressing global health challenge.

Introduction

Obesity is one of the main health problems affecting children. Childhood overweight and obesity of children will increase the probability of adult obesity and affect children's mental health development, potentially inducing chronic diseases such as diabetes, hypertension, cardiovascular disease and even certain tumors. 1–3 Many factors are related to childhood obesity, and excessive screen time has received increasing attention in recent years. 4–6 As digital technology advances rapidly, children's daily routines are increasingly dominated by screen-based activities. The ubiquity of devices like smartphones and tablets has intensified concerns that excessive screen exposure negatively impacts children's health, with obesity and overweight being key worries.

For the purposes of this review, screen time encompasses time spent on all screen-based devices, including televisions, computers, smartphones, tablets, and handheld gaming consoles. It is

important to distinguish between recreational screen time (eg, television viewing, video gaming, social media use) and educational or interactive screen use, as these modalities may exert differential effects on health outcomes. Furthermore, screen-based activities are heterogeneous, passive consumption of entertainment content, active engagement with educational software, and interactive social media participation likely operate through distinct behavioral and psychological pathways, and treating them as a uniform exposure may oversimplify the existing evidence.

Observational studies have consistently reported that prolonged screen exposure is associated with elevated childhood obesity risk, and this relationship has become a public health priority. 4 For example, children exceeding two hours of daily screen time face significantly higher odds of overweight/obesity than peers with under two hours' exposure. 4 This association appears to stem from several interconnected pathways, screen-based activities may be associated with less time available for physical activity and with eating or snacking in a distracted context; 5 prolonged sedentary periods generally involve lower energy expenditure than many forms of active behavior and may, in some contexts, contribute to an imbalance between energy intake and expenditure; 7 and blue light exposure may delays melatonin release, fragmenting sleep patterns. 5 , 8 Content context further modifies outcomes, educational programming occasionally supports healthy habits, yet pervasive junk food marketing drives poor dietary choices. 9 , 10 Parental mediation strategies such as co-viewing discussions, can mitigate risks by shaping consumption awareness. 9 In addition, chronic screen engagement has been associated with heightened anxiety and depression, 8 which may trigger emotional eating and activity avoidance, thereby reinforcing a cycle of unhealthy behaviors. Nevertheless, it is critical to note that much of this evidence is observational.

Screen use patterns, developmental needs, and susceptibility to obesity-related factors differ considerably across developmental stages. Preschool children predominantly consume passive content and are highly influenced by parental co-viewing practices; school-aged children begin engaging with interactive educational and gaming content; and adolescents demonstrate high autonomy in social media and streaming consumption, often accompanied by late-night use that disrupts sleep. 11–13 These age-related differences suggest that the relationship between screen exposure and adiposity is not uniform across childhood. Socioeconomic constraints, cultural norms around technology use, and parental mediation practices all affect children's screen engagement patterns, 14 with significant implications for physical and psychological well-being.

Despite extensive research, several gaps remain. The relative contributions of different screen modalities to weight gain are incompletely understood. Moreover, the extent to which screen time operates independently of confounding lifestyle factors, including dietary patterns, baseline physical activity, sleep duration, socioeconomic status, and family environment, requires clearer elucidation. The present narrative review addresses these gaps by synthesizing evidence across screen types, age groups, and mechanistic pathways, with explicit attention to study design limitations and the distinction between association and causation.

Compared with previous reviews that have often focused primarily on total screen duration or on a single behavioral pathway, this review integrates evidence across screen modality, content, developmental stage, and viewing context. It also places greater emphasis on the distinction between observational association and causal inference, evaluates proposed behavioral, dietary, sleep-related, and psychological pathways separately, and considers how these distinctions may inform targeted interventions rather than uniform screen-time reduction. Effective public health strategies must account for household resources, caregiver attitudes, and community environments that shape children's media experiences, moving beyond simple screen-time limits to promote

healthy lifestyles and obesity prevention.

Classification of Screen Time and Its Associations with on Children Health

Definition and Classification of Daily Screen Time

Broad Classification of Daily Screen Time for Children

The daily screen time can be divided into several types: passive entertainment (television viewing, streaming), interactive content such as active video games, social media engagement, and educational software. 15 , 16 Educational programs often appear on dedicated platforms and are becoming increasingly popular among children and adolescents, serving as a medium for learning various subjects. 17 , 18 Recreational videos cover various types from cartoons to music videos, primarily aimed at attracting viewers in fun and relaxing ways. Social media platforms featuring short videos are particularly attractive to young audiences due to their fast-paced consumption format. Recent studies have shown that children aged 3–4 watch an average of 61.2 minutes per day on mobile YouTube, with children from lower socioeconomic backgrounds watching for longer periods of time. 11 It is essential to recognize that these categories are not mutually exclusive and may exert differential effects on energy balance behaviors.

Age-Related Differences in Screen Use Patterns

Age plays an important role in the type and duration of video content consumed. Older children and adolescents tend to engage more with interactive content such as video games and social media platforms, whereas younger children are more likely to consume passive content such as cartoons and educational videos. A study on preschool children found that mobile video websites and apps are commonly used by parents with lower levels of education, indicating a potential correlation between socioeconomic factors and screen consumption habits 11 During the COVID-19 pandemic, increased use of video chat and social media among adolescents led to delayed sleep time, which indicates that the nature of video content consumed can affect daily life and behavior. 12 This age-related difference highlights the importance of considering developmental stage when evaluating screen consumption patterns.

Screen Content, Context, and Child Health

The content, quality, and context of children’s screen use may be associated with their social and emotional development and overall well-being. These associations may vary according to the type of content, developmental stage, duration and timing of exposure, and degree of parental mediation. High-quality educational content, especially content that promotes social connections, is related to children’s positive social behavior. Conversely, excessive exposure to harmful or low-quality content may lead to negative outcomes, such as increased problematic media use and shortened sleep time. 19 , 20 For example, children who view content promoting unhealthy eating behaviors may experience negative body image concerns and unhealthy eating habits. 21 , 22 In addition, the rise of certain video platforms that often feature food related content has raised concerns about the media’s perception of young audiences’ body image and dietary habits. Therefore, parents and educators must therefore consider content quality and viewing context, rather than focusing solely on duration.

The Relationship Between Daily Screen Time and Children’s Weight

The rise in childhood obesity is multifactorial, but the modern sedentary lifestyle, characterized by increased screen time, reduced exercise, and poor diet, is a major contributor.

Changes in Body Mass Index

Observational studies have examined the relationship between daily screen time and children's body mass index (BMI), which has attracted widespread attention. Meta-analytic evidence indicates that increasing screen time, especially exceeding two hours per day, is associated with higher BMI in children. 23 , 24 For example, a study found that children who spend more time on screens are much more likely to be overweight or obese compared to those with limited screen time. 25 Specifically, the likelihood of being overweight may increase with every hour of screen time, highlighting a concerning trend in children's health as screen time increases.

Nagata et al reported that for every 1-hour increase in screen time, the BMI percentile increases by 0.22. 23 In addition, considering the sedentary nature of screen activity, these effects may be compounded by decrease physical activity levels that contribute to weight gain. This association emphasizes the need for parents and caregivers to effectively monitor and manage children's screen time to promote healthier weight outcomes and overall well-being. 26 , 27 Compared with those who use entertainment electronic screens for less than 4 hours per day, adolescents who use entertainment electronic screens for 4–8 hours and>8 hours per day showed a 24% and 29% increased risk of overweight or obesity, respectively. 28 Nevertheless, these estimates rely on self-reported screen time, which is prone to measurement error and social desirability bias. However, this relationship is not universally observed. Some recent longitudinal evidence reported weak, mixed or null associations after adjusting for baseline physical activity, dietary intake, and parental BMI, suggesting that screen time may partly reflect broader lifestyle patterns and family-level characteristics rather than acting as an independent determinant of adiposity. 29–31 Furthermore, effect sizes may differ by screen modality television viewing tends to show stronger and more consistent associations with adiposity than active video gaming or educational screen use, whereas the effects of social media and mobile-device use may depend on their influence on sleep, diet, and physical activity. 29 , 30 As summarized in Table 1 , several cited studies reported higher BMI-related measures or higher odds of obesity among children with greater screen exposure. However, differences in exposure categories, study populations, outcome definitions, and adjustment strategies limit direct comparison across studies and do not establish a causal dose–response relationship. Table 1 Associations Between Daily Screen Time and Weight-Related Outcomes in Children Daily Screen Time Reported BMI Z-Score Estimate Reported OR for Obesity References <1 hour −0.12 (95% CI: −0.20, −0.04) 1.00 (Ref) [23 , 24] 1–2 hours +0.18 (95% CI: +0.05, +0.31) 1.45 (1.22–1.72) [23 , 24] 2–4 hours +0.41 (95% CI: +0.29, +0.53) 1.89 (1.65–2.17) [4 , 23 , 24] >4 hours +0.73 (95% CI: +0.61, +0.85) 2.54 (2.20–2.93) [23 , 31] Notes : BMI Z-score changes are presented as point estimates with 95% confidence intervals. The studies may differ in age range, sample characteristics, screen-time assessment, outcome definitions, covariate adjustment, and statistical methods. Abbreviations : OR, odds ratio; CI, confidence interval; Ref, reference category.

Associations may also differ by sex, although the available evidence remains inconsistent. Differences in social media use, gaming patterns, physical activity, dietary behaviors, and body-image concerns may partly contribute to sex-related variation in the observed associations.

Trends in Overweight and Obesity

In recent years, the incidence of overweight and obesity in children has sharply increased, and screen time has emerged as an important associated factor (Table 2). Data from various studies indicate a sharp increase in overweight and obesity rates among adolescents. 32 , 33 Global trend analyses show that obesity rates are significantly increasing in many countries, especially among those who overuse screens. 33 , 34 These trends are of public health concern because childhood obesity is associated with an increased risk of several adverse health outcomes, including type 2 diabetes, cardiovascular disease, and psychosocial difficulties. 35 , 36 A survey involving 5180 children in South Korea showed that prolonged screen time on smartphones and televisions is closely associated with higher obesity rates, while reducing one hour of screen time while increasing one hour of non-screen time is significantly associated with lower obesity rates. 37 However, isotemporal substitution analyses from the same study suggested that the benefit of replacing screen time with physical activity was more pronounced for television viewing than for smartphone use, underscoring the heterogeneity of screen modalities. 2.3.3 Epidemiological evidence and illustrative studies. Table 2 Screen Time Modalities and Reported Health-Related Associations Category Examples Average Daily Use Associated Outcomes References Passive Entertainment Television, streaming video 2.8 hours Higher BMI and poorer sleep quality [5 , 38–40] Interactive Media Video games, virtual reality 1.5 hours Higher obesity risk and poorer executive function [20 , 41 , 42] Social Media Short-video platforms, image-based social networking 2.1 hours Increased emotional eating and greater body dissatisfaction [10 , 21] Educational Online learning, instructional software 1.2 hours No apparent effect on weight when use is less than 2 hours per day [17 , 18] Abbreviations : BMI, body mass index; OR, odds ratio; R, Pearson correlation coefficient.

Epidemiological data on screen time and weight-related outcomes provide further evidence of the public health relevance of this issue. Children who participate in more than two hours of daily screen time are at approximately 1.5 times greater odds of overweight than their peers who have less screen exposure. 4 , 43 However, considerable heterogeneity has been reported across studies, partly because of differences in age groups, study design, screen-time assessment, and adjustment for confounding factors. Findings may also vary across geographic and cultural settings because screen-use norms, parental mediation practices, school schedules, and opportunities for physical activity differ between countries and regions. Differences in the definition and measurement of screen time, access to digital devices, and family media-use rules may contribute to cross-regional variation in the observed associations.

In a study conducted in Hong Kong, researchers found that the proportion of children watching TV every day has decreased, while the use of electronic games has significantly increased, which is related to the rise in obesity rates. 27 This dichotomy emphasizes the necessity of targeted interventions that focus on the types of screen time children participate in, rather than just the duration. Television viewing, gaming, social media use, educational screen use, and mobile-device use may have different effects on weight-related behaviors. Effective strategies may include promoting educational screen time and encouraging physical activity breaks during prolonged screen use, thereby reducing the adverse effects of excessive screen time on children’s weight. 44 , 45 Not all longitudinal studies have reported significant prospective associations between screen exposure and subsequent BMI or BMI z-score. In particular, associations may attenuate or disappear after accounting for baseline adiposity, physical activity, dietary intake, and family-level factors. 29 , 30 These inconsistent findings suggest that screen time may be both a potential risk factor and a marker of broader lifestyle and family conditions. Therefore, the relationship should be interpreted according to screen modality and context rather than total screen duration alone.

Proposed Pathways Linking Screen Time to Weight-Related Outcomes

Excessive daily screen time in children (≥ 2 hours) has been associated with overweight and obesity. This association may involve several interrelated behavioral and contextual factors, including reduced physical activity, greater sedentary time, unhealthy eating behaviors, psychological difficulties, and sleep disruption. However, these factors should be considered potential or proposed pathways rather than established causal mechanisms, and their independent contributions remain difficult to quantify.

Reduced Physical Activity

Lack of regular physical activity and sedentary lifestyle are one of the most important determinants of childhood obesity. Screen-based activities often lead to prolonged sitting and reduced energy expenditure. The decrease in activity is a well-documented factor leading to various health problems, including obesity, cardiovascular disease, and mental health disorders. Research has shown that sedentary behavior, such as prolonged video viewing, characterized by prolonged inactivity, is associated with adverse health outcomes. A study in Germany showed that more than 2 hours of screen time per day significantly reduces physical activity time and increases sedentary time, which may be one of the important contributors to overweight and obesity in children. 46 Jones et al found that children who spend more than 2 hours of screen time per day can significantly reduce their exercise time, demonstrate markedly lower physical activity levels, thereby increasing the risk of sedentary behavior and overweight/obesity. 47 In addition, during the COVID-19 pandemic, screen time became one of the main ways of teaching and learning, which aggravated the sedentary lifestyle, because many people spent more time at home, resulting in reduced physical activity and increased sedentary behavior. 25 , 48 A study on asymptomatic peripheral arterial disease patients showed that a 12 week intervention aimed at reducing sedentary behavior significantly improved physical activity levels and endothelial function, highlighting the importance of lifestyle changes in managing health risks. 49 The longitudinal effects of reducing physical activity can be manifested in both physical and psychological domains, and there is evidence to suggest that even moderate increases in daily activities can produce significant health benefits. 50 This emphasizes the necessity of enhancing physical activity interventions for children with excessive screen time to mitigate the health risks associated with prolonged screen time and related lifestyles.

Changes in Dietary Habits

The association between increased screen time and weight gain may be related to concurrent increasing food intake. Numerous studies have shown that this association is evident across multiple screen-based activities: watching television boosts positive responses toward food intake and snacking behaviors in children and adolescents, 38–40 with similar effects observed in relation to electronic games 41 , 42 and personal computer use. 51 Food-related advertisements in screen content may affect children’s dietary behavior by promoting the consumption of junk food and fast food, as well as increasing obesity. 52–54 Jones et al found that excessive screen time per day in children can be associated with poor dietary habits, and excessive screen time can cause children to consume excessive high calorie food. 46 , 47 It is important to acknowledge that the evidence linking specific screen content to dietary intake is predominantly observational, and the influence of concurrent parental dietary habits and household food availability represents a significant confounding factor that is not always adequately controlled. Confinement measures simultaneously triggered both dietary polarizations, stress-induced consumption of energy-dense snacks/fast food 55 and health-conscious adoptions of plant-based patterns, 56 and unprecedented screen-time surges through virtual schooling and social displacement. Children’s prolonged screen engagement may

enables mindless snacking during viewing sessions while normalizing food marketing exposure, and parental stress from disrupted routines often manifests in permissive screen and dietary practices. 57 This screen-diet nexus operates through cognitive pathways where stress/impaired emotional regulation concurrently drive comfort-food seeking and screen-mediated escapism in children. Understanding these codependent patterns is essential for designing integrated interventions that address screen habits and nutritional outcomes synergistically during public health crises.

Sleep Disruption

Prolonged screen exposure, particularly near bedtime, has been associated with later sleep timing, longer sleep-onset latency, and shorter or poorer-quality sleep in children and adolescents. These associations may involve both behavioral displacement and light-mediated circadian mechanisms, although direct evidence linking routine screen exposure with melatonin suppression in children remains limited. 58 , 59 Short sleep duration is prospectively associated with weight gain in pediatric populations, potentially via alterations in leptin and ghrelin signaling that increase hunger and reduce satiety. 60 However, the independent contribution of screen-related sleep disruption to childhood adiposity remains difficult to quantify because screen exposure, sleep duration, physical activity, and dietary behaviors are interdependent and may share common social and family-level determinants. 60 , 61

Psychological Health Factors

In recent years, with the widespread use of electronic devices worldwide, the relationship between excessive screen time and pediatric mental health. Multiple studies have shown that excessive screen time is associated with various problems such as impaired emotional development and mental health in young children, including anxiety, depression, emotional instability, and poor attention, which in turn can affect their daily routines and dietary habits and may lead to obesity and reduced physical activity. 62–64 Excessive screen time affects children’s psychological well-being, which in turn influences obesity risk. Children with anxiety or depression may spend more time on screens, where depressive symptoms are associated with reduced outdoor play and increased compulsive scrolling. This pattern creates a self-perpetuating cycle: increased screen dependency reduces engagement in physical activities like team sports, while blue light exposure from late-night viewing worsens emotional symptoms (Figure 1). 65 However, causal inference is limited by the cross-sectional nature of much of this research, and it remains unclear whether screen time precipitates mental health difficulties or whether pre-existing vulnerabilities drive higher screen engagement. Figure 1 Mediating mechanisms underlying screen time and childhood overweight/obesity. Prolonged screen exposure (>2 h/day) contributes to childhood overweight/obesity via four primary pathways shown in the diagram. Bold■formatted text boxes represent major pathway headings. The flowchart illustrates how excessive screen time, defined as more than 2 hours per day, leads to childhood overweight or obesity through four pathways: behavioral, dietary, sleep and psychological. The behavioral pathway shows that increased sedentary behavior and decreased physical activity reduce energy expenditure. The dietary pathway indicates that increased food marketing exposure and mindless snacking raise caloric intake. The sleep pathway describes how increased blue-light exposure suppresses melatonin secretion, leading to decreased sleep duration and quality. The psychological pathway explains that increased anxiety and depressive symptoms, along with emotional instability, contribute to emotional eating. All pathways converge to result in childhood overweight or obesity. A flowchart showing pathways from excessive screen time to childhood overweight/obesity.

Breaking this cycle requires addressing both mental health and screen habits. Some evidence suggests that resilience-building programs teaching emotion-regulation techniques are associated with reduced recreational device use and healthier snacking patterns, 66 indicating that psychological support may help modify digital consumption behaviors. Therefore, effective screen time mitigation requires dual-pathway strategies: reducing device access combined with alternative activities such as nature exposure or creative play. Screen-limiting measures alone may be insufficient without addressing underlying emotional needs.

Intervention Strategies and Recommendations

Limiting Daily Screen Time and Promoting Alternative Activities

The increasing use of screen-based devices, particularly among children, has raised concerns about their potential associations with physical and mental health outcomes. Restricting daily video gaming and screen exposure thus emerges as a crucial measure for promoting healthier lifestyle choices.

Research consistently links excessive screen time with various adverse health outcomes, including obesity, sleep disturbances, and mental health issues such as anxiety and depression. 67 For example, a study in China observed that adolescents complying with new gaming time restrictions shifted their leisure activities: while some turned to healthier alternatives like sports, there was also a concerning rise in short video consumption. 68 This implies that simply limiting gaming duration, while potentially reducing that specific activity, may not effectively decrease overall sedentary time and could even trigger a shift towards alternative screen dependencies.

Consequently, restrictive measures must be paired with proactive strategies that actively encourage participation in physical exercise and hobbies beneficial to well-being. Policymakers need to integrate educational components that clearly communicate the risks of excessive screen use to adolescents and offer practical alternative activities, fostering more active and engaged lifestyles.

Promoting Active Lifestyles and Healthy Habits

Reducing sedentary behaviors and promoting active lifestyles require parental leadership as a foundation for behavioral change. Parents should reduce their own screen time and engage preschoolers in shared outdoor activities such as family walks, park visits, and nature exploration to increase active outdoor time. 69 This approach addresses directly addresses the sedentary patterns exacerbated by excessive screen use and modern conveniences, as encouraging active lifestyles serves as a critical counterbalance to sedentary behaviors.

Healthy dietary habits should also be encouraged. Parents can model behavior by reducing consumption of fried foods, high-fat snacks, and sugar-sweetened beverages, and by establishing regular meal/snack times with whole foods. Regular physical activity improves children’s physical health and mental well-being, as was evident during the COVID-19 pandemic when activity levels declined. 70 , 71 Effective strategies therefore require multifaceted collaboration: community resources like accessible parks and recreational facilities must support family activities, while schools and public health campaigns should synchronize messaging to reinforce parental leadership. Crucially, programs facilitating parent-child outdoor engagement demonstrate compound benefits, improving physical health, strengthening family bonds, enhancing social competence, and deepening community connections, ultimately demanding an integrated approach that empowers parents as daily exemplars and active partners in embedding healthy movement and eating routines within early childhood development directly linking the reduction of sedentary behaviors to the cultivation of active

lifestyles.

A Comprehensive Strategy for Intervening in Childhood Obesity Related to Excessive Screen Time

In order to reduce overweight and obesity in children caused by screen time, evidence-based interventions must involve teachers and communities as frontline responders. 47 , 72 Teachers can integrate dynamic exercise breaks during classroom transitions, clearly simulating device free leisure habits, and embed nutritional literacy into science courses, using interactive courses to demonstrate how excessive snacking during screen use can lead to weight gain. 73 At the same time, communities should activate underutilized spaces: transform the library courtyard into an after-school jumping house area, collaborate with pediatricians to distribute screenless challenge toolkits at local clinics, and organize weekend family sports festivals to replace fried food vendors with cooking demonstrations of vegetable snacks. It is crucial to share and track children's non screen activity time through a simple reward system such as tokens redeemable for amusement park equipment, synchronize the efforts of the school community, and ensure sustained reinforcement of healthy behavior in all environments where children live and learn. In addition, policymakers can consider regulations such as limiting certain electronic games for minors and reducing non-essential screen use in schools.

Conclusion

This review examined the association between daily screen time and childhood overweight and obesity, highlighting the importance of moderating excessive recreational screen exposure. Evidence indicates that increased screen time is generally associated with higher overweight and obesity rates in children. However, the magnitude and consistency of this association vary according to screen modality, content, developmental stage, measurement method, and family or social context. Television viewing has shown more consistent associations with adiposity in some studies, whereas findings for active video gaming, educational screen use, and handheld-device use have been less consistent. Therefore, effective interventions are needed to reduce excessive and sedentary screen use while promoting healthier daily routines. Although many studies report adverse associations between screen exposure and weight-related outcomes, the available evidence is predominantly observational.

The relative contribution of different mechanisms also remains uncertain. Sedentary behavior, displacement of physical activity, exposure to unhealthy food marketing, snacking during screen use, and sleep disruption should therefore be regarded as potential or proposed pathways rather than definitively established causal mechanisms. Clarifying these mechanisms will help guide future intervention design.

Given the multifaceted nature of childhood obesity, intervention measures should address multiple levels, reducing excessive recreational screen time, promoting regular physical activity, supporting adequate sleep, and encouraging healthy eating habits. Parental guidance on screen content and viewing routines may also improve outcomes, particularly among younger children. Current evidence suggests that screen exposure may influence children's health through several interrelated behavioral pathways, including reduced physical activity, increased sedentary time, dietary changes, and delayed or shortened sleep. However, the independent contribution of each pathway, after accounting for baseline adiposity, diet, physical activity, sleep, socioeconomic conditions, and family lifestyle, remains difficult to quantify. Policy measures and community programs addressing these

broader lifestyle and environmental factors may create healthier environments.

In summary, screen time should be considered one potentially modifiable component of the broader obesogenic environment rather than an isolated cause of childhood overweight and obesity. Collaborative efforts across families, educational institutions, healthcare systems, and policymakers can support children’s physical and mental health and promote balanced digital-media use alongside active lifestyles, healthy diets, and adequate sleep (Figure 2). Future research should prioritize well-designed longitudinal studies using objective or device-based measures of screen exposure, detailed classification of screen modality and content, repeated assessment of adiposity and relevant behavioral mediators, and comprehensive adjustment for confounding factors. Well-powered randomized controlled trials and pragmatic intervention studies are also needed to determine whether reducing specific types of screen use produces clinically meaningful improvements in BMI, adiposity, sleep, physical activity, or dietary behavior, and to identify effective interventions for different age groups and social contexts. Figure 2 Schematic overview of screen time related risk for childhood overweight and obesity. This conceptual framework illustrates the multi■step association between screen time and pediatric overweight/obesity. Bold■colored boxes represent grouped conceptual blocks. A flowchart depicting the progression from background factors to overweight/obesity in children. The first section, Background factors, includes socioeconomic status, cultural influences and parental perceptions and practices. An arrow leads to Screen exposure, which lists screen time, content type and parental mediation. Another arrow points to Impact on the body, detailing reduced physical activity, changes in dietary behavior and impact on sleep/psychology. This connects to Physiological outcomes, showing reduced energy expenditure, increased energy intake and circadian rhythm disorder. An arrow leads to Disrupted energy balance, which finally connects to Overweight/obesity in children. Flowchart of factors causing childhood obesity, from background to physiological effects.